Sipeng Xie

(+86) 133-5084-1699 | sipengxie@buaa.edu.cn | sipengxie2024.github.io | GitHub

Systems research & engineering | Compiler/runtime · Parallel execution · Storage · AI infrastructure



Education

Beihang University, Beijing **2022 – Jun. 2027 (Expected)**
Ph.D. in Cybersecurity, advised by Prof. Qianhong Wu

Beihang University, Beijing **2021 – 2022**
Graduate study in Cyberspace Security

Beihang University, Beijing **Sep. 2017 – Jul. 2021**
B.E. in Information Security; second major in Mathematics. GPA 3.7625/4.0; Mathematics 95.33/100

Chengdu No. 7 High School (Linyin Campus) *NOIP Second Prize, 2014 and 2015* **Sep. 2014 – Jun. 2017**

Publications

- **Sipeng Xie**, Qianhong Wu, et al. *SkelOT: Reusing AOT Compilation Across EVM Contract Families.* **EuroSys 2027 (accepted).**
- **Sipeng Xie**, Qianhong Wu, et al. *MHOT: Height-Optimized Authenticated Data Structure for Blockchain State Commitment.* **USENIX Security 2026.**
- **Sipeng Xie**, Qianhong Wu, et al. *VAM: Client-Verifiable State Continuity for Outsourced Agent Memory.* **NDSS 2027, under review (second round).**
- **Sipeng Xie**, Qianhong Wu, et al. *The Proxy Knows Too Much: Sealing LLM API Routers with Attested TEEs.* **IEEE TIFS, under review.**
- Wenkuan Xiao, Qianhong Wu, Wenbo Wu, **Sipeng Xie**, et al. *AccountCatcher: Anomaly Blockchain Account Detection Based on Hybrid Graph-Based Model.* **ProvSec 2025.**
- Tianxu Han, Qianhong Wu, Kun Wang, Yingmiao Zhang, **Sipeng Xie**. *Proof of Inaction for Outsourced Data with Exhaustive Audit.* **Computers & Electrical Engineering 2024.**
- Tianxu Han, Qianhong Wu, **Sipeng Xie**, et al. *Iterative Proof of Retrievability for Periodic Incremental Data via Optimistic Tags Generation.* **CSCloud 2024.**
- Kun Wang, Qianhong Wu, Tianxu Han, **Sipeng Xie**, et al. *PDTS: Practical Data Trading Scheme in Distributed Environments.* **ProvSec 2024.**
- Tianxu Han, Jian Mao, **Sipeng Xie**, et al. *VM-Studio: A Universal Crosschain Smart Contract Verification and Execution Scheme.* **Security and Communication Networks 2023.**

Engineering Experience

paradigmxyz/revmc EVM JIT/AOT compiler on LLVM **5 merged PRs · 2026**
LLVM UB fix in guarded binops; i256 as i64 lanes with tighter alias attributes; ruint DIV/MOD; `resume_at` persistence; revm v36.

Wei-Shaw/sub2api LLM API gateway **2 merged, 7 open PRs; 3 issues · 2026**
Merged: web_search emulation on `/v1/messages`, verbatim OAuth image prompts. Open: speech-to-text, DeepSeek web search.

Cryptography and distributed systems **Merged PRs and issues · 2026**
tlsnotary/tlsn: size MPC-TLS GHASH preprocessing to the record length (merged). succinctlabs/flock: pooled Merkle-path inversions, $4 \rightarrow 1$ (merged). tempoxyz/tempo: reported quadratic payload-build rescans. temporalio/temporal: start-time rounding fix (open).

Blockchain Execution Infrastructure · $11M Pre-Seed (confidential) **2024 – 2026**

Ant Group · Ant Star Internship **Ongoing**
Ant Digital Technologies, Digital Technology Division

AI and Developer Infrastructure

DeepSeek-V4.1-Flash Inference Optimization vLLM, 8× RTX PRO 6000 over PCIe **2026**
SM120 FP4 tensor-core indexer; two-half prefill micro-batching that hides PCIe all-reduces; copy-engine all-reduce (21.4 → 28.3 GB/s) with 2 SMs reserved for MoE. 790K-token cold start: prefill 171 → 49.5 µs/token (3.5×; TTFT 135 → 39 s); decode +43–50%; accuracy within ±1 pp. Indexer kernels upstreamed in 2 merged PRs (yhfgyyf/vllm-deepseek-v4-sm89).

Pi Agent Runtime Learning Journal **Open source · 2026**

Laboratory AI Router **Beihang University · Ongoing**

Technical Skills

Languages & tools: Rust, Python; working knowledge of C/C++ and TypeScript; LLVM, Linux, Git, code review.
Systems: Compiler/runtime optimization, parallel execution, authenticated storage, benchmarking.